

SERGIO KEVIN PEREZ NATEROS

Backend Engineer | Python · FastAPI · Django

Healthcare Systems Integration · Process Automation

Tingo Maria, Peru • Remote worldwide • UTC-5 (full-day overlap with US time zones)

sperezn.dev@gmail.com • skapir.github.io • github.com/Skapir • linkedin.com/in/sergio-perez-nateros

PROFESSIONAL SUMMARY

Backend engineer with 5 years building production Python systems on top of 13 years inside Peru's public health system. I specialize in integrating legacy healthcare platforms with no public API — HTTP reverse engineering, workflow automation, and data pipelines that eliminated manual clinical-administrative processes for a high-demand hospital. Core stack: Python (FastAPI, Django), PostgreSQL, Celery and Playwright. End-to-end ownership: requirements, architecture, implementation, deployment and end-user support.

PROFESSIONAL EXPERIENCE

EsSalud – Hospital I Tingo Maria (Peru's National Social Health Insurance)

Apr 2013 – Present

Tingo Maria, Peru — public hospital network serving the Huanuco region

Backend Developer / Systems Analyst – Patient Referrals Unit

2021 – Present

- Automated the end-to-end medical referral workflow on top of closed institutional platforms with no public API, by analyzing HAR captures and reverse-engineering their HTTP traffic (persistent sessions with keep-alive pings): the system pulls referrals issued by physicians, consolidates them for the referral officer's approval, and flags inconsistencies in clinical notes and ICD-10 coding before submission — cutting rejected records and rework between facilities.
- Own the rejected-referral remediation cycle across multiple medical specialties, including routing decisions between hospitals of different care levels, keeping patient continuity of care intact.
- Designed and shipped internal Python automation systems now running in production, replacing manual processes previously handled by clinical and administrative staff.
- Processed 15 months of historical referral and physician-scheduling data programmatically: the resulting reports supported the CT scanner procurement case before management, exposed a 399% demand-to-capacity gap in General Medicine, and proved the facility's elevated appointment-deferral indicator was driven by follow-ups rather than new demand — correcting the institutional metric.
- Automated executive reporting (Excel, Word, PowerPoint) with Python, replacing manually assembled monthly and quarterly technical reports.

Healthcare Data & Operations Assistant – Multi-department rotation

2013 – 2021

- 8 years operating the hospital's clinical-administrative systems (ESSI, REFCON, SGSS) across every department — the domain knowledge the facility's automation systems were later built on.

Independent Software Developer & Consultant

2024 – Present

Selected consulting for healthcare and retail clients

Remote

- Pharmacy point-of-sale and inventory platform featuring national drug-registry lookup, cash management, customer loyalty, and FEFO lot tracking with expiry control.
- Middleware layer allowing an HL7-based clinic to query and operate an AMIS-850 medication dispenser that ships with a proprietary SDK only and no native HL7 support.
- FotoStudio Pro: e-commerce for a professional photography studio with watermarked catalog, shopping cart, mobile-wallet (Yape) payment and digital delivery of high-resolution photos.

SELECTED PROJECTS

Full write-ups for every project at skapir.github.io

CitaExpress — Medical referral delivery middleware

FastAPI, SQLite, Jinja2, HTMX, TailwindCSS, ReportLab, httpx

Production middleware that scrapes scheduled referrals from a closed hospital platform, generates stamped PDFs and delivers them to patients over WhatsApp links, removing unnecessary trips to the hospital. Built on a persistent httpx session with keep-alive pings to avoid lockouts on the source system.

PediaClinic — Pediatric clinic management platform

Django 4.2, PostgreSQL, Celery, HTMX, Tailwind/Flowbite, ReportLab

Role-based platform for guardians, technicians, physicians and admins, with dual authentication (Google OAuth2 and internal credentials), mobile-wallet payments reconciled automatically over IMAP, reservation expiry via Celery Beat, Telegram bot notifications and PDF receipts.

HL7 Middleware — AMIS-850 medication dispenser

Python, HL7, proprietary SDK

Clinical interoperability layer that translates HL7 messaging into the proprietary SDK calls of an AMIS-850 medication dispenser, letting a clinic operate it from its standard infrastructure with no native vendor support.

TECHNICAL SKILLS

Languages: Python, SQL, JavaScript, HTML5, CSS3

Backend: FastAPI, Django, Django REST Framework, SQLAlchemy, Celery, httpx

Frontend: HTMX, Jinja2, TailwindCSS, Flowbite, Bootstrap

Databases: PostgreSQL, SQLite

Automation: Playwright, web scraping, HAR analysis and API reverse-engineering, ReportLab, openpyxl

Data: Pandas, Power BI, statistical analysis, automated report generation

Integrations: HL7, OAuth2, IMAP/SMTP, REST APIs, Telegram bots, WhatsApp

Tools: Git, GitHub, Docker, GNU/Linux

EDUCATION

B.Eng. in Systems & Computer Engineering — Universidad Tecnologica del Peru (UTP) 2021

Professional Engineering License (Titulo Profesional), Peru 2026

CERTIFICATIONS (SELECTED)

Backend Development with Python track — Platzi (2025) · Django REST Framework — Platzi (2024) · Python Developer, 220 hrs — Edutin Academy (2025)

LANGUAGES

Spanish — Native · **English** — Intermediate (B1): technical reading and written communication; comfortable with async, written-first collaboration